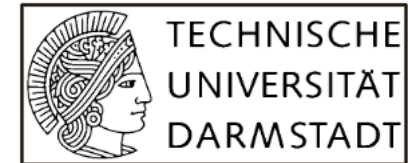
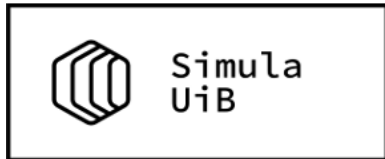


Counter Galois Onion (CGO)

A New Onion Encryption Scheme for Tor

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GAPS Workshop, Vienna 2026



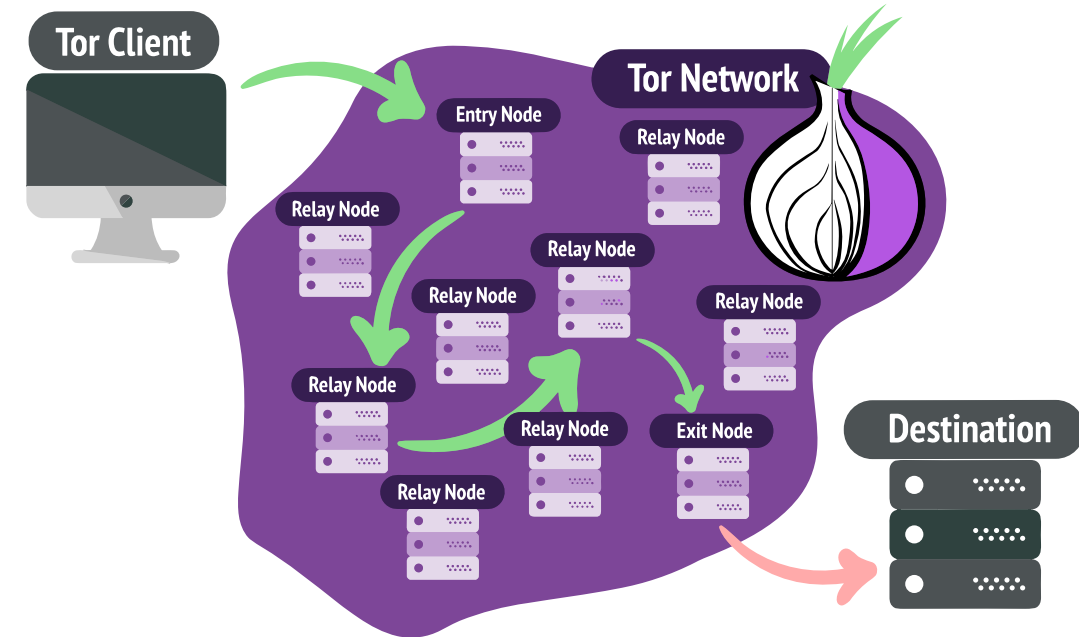
Outline

- Tor Background
- Historical Timeline of Evolving Solutions
- Design of CGO and its Rationale

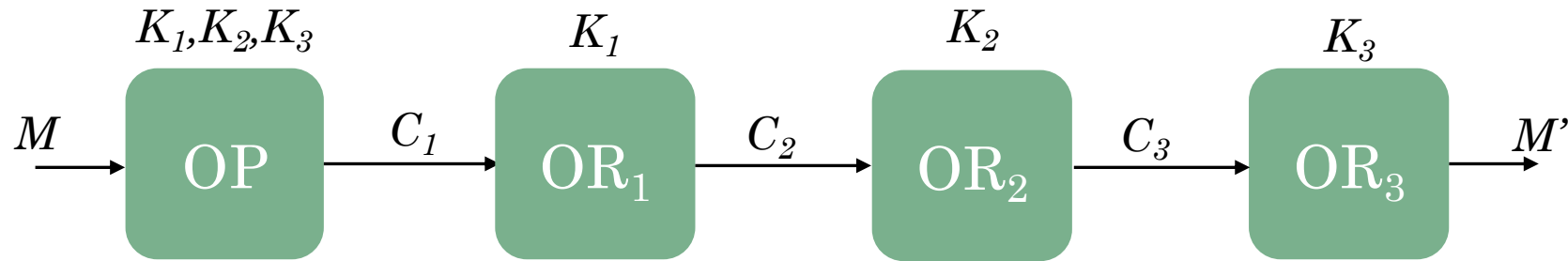
Tor's Onion Encryption and its Limitations

Tor

- Tor is the most popular system for maintaining **anonymity on the web**, first described in a 2004 paper by **Dingledine, Matthewson**, and **Syverson**.
- It consists of a network of Onion Routers (**OR**), mostly operated by volunteers, and an Onion Proxy (**OP**) operated by the user and typically incorporated as part of a browser (**Tor browser, Brave**).
- It employs multilayered encryption called **onion encryption**.



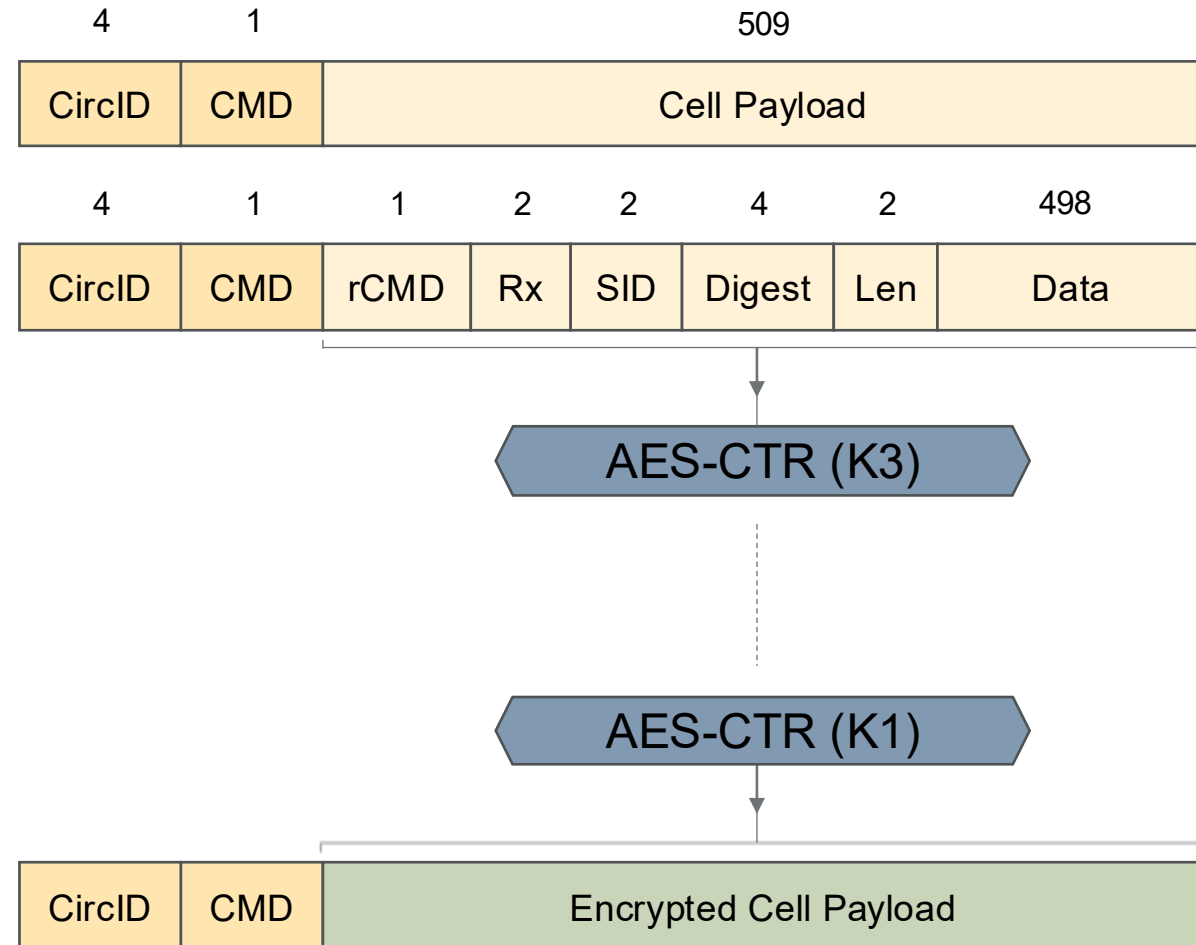
Onion Encryption in Tor



- In Tor, data flows through **circuits** consisting of an **Onion Proxy (OP)** and a sequence of **Onion Routers (OR)**.
- Circuits are **analogues of symmetric channels**, where the Onion Proxy shares a symmetric key with each Onion Router.
- Each Onion Router “**peels off**” a layer of encryption, **decorrelating incoming ciphertexts** from **outgoing ciphertexts**.

Relay Cell Format and Processing [DMS04]

- Cells are 514 bytes (v4+)
- **CirclD**: Circuit Identifier
- **CMD**: Cell type - RELAY (3) or RELAY_EARLY (9)
- **Rx**: Recognised field (0x0000)
- **Digest**: seeded running hash (truncated SHA-1)



Notable Properties

- It allows “**Leaky Pipes**”, meaning that the **OP** can send a message to **any OR** in the circuit, **and vice versa**.
- An **OR** either recognizes a cell (via **Rx & Digest**), meaning it is the intended recipient, **or forwards the cell** further down the circuit.
- The **malleability of AES-CTR** propagates through **multiple layers**, enabling a **de-anonymizing** attack called a **tagging attack**.
- **Bits** are flipped in a ciphertext at **OR₁** and reversed at **OR₃** to **test** whether **decryption succeeds** $\Rightarrow$ ciphertexts belong to the **same circuit**.

Historical Timeline of Evolving Solutions

[2012] Tor Proposal 202

- In **Proposal 202** Nick Matthewson, expressed the need to update the onion encryption in Tor.
- Specifically, it called for Onion Encryption that:
 1. Protects against **Tagging Attacks**,
 2. Is **non-expanding**, so as not to leak circuit size and relative position,
 3. Offers **Stronger end-to-end integrity** protection,
 4. Does not use **MAC-then-Encrypt**.

- In **Prop 261(2016)** Matthewson proposed a new OE scheme where each encryption layer consists of a **Wide-Input Tweakable Cipher**.
- Intuitively, **WITC is non-expanding, non-malleable, and integrity can be achieved by adding redundancy to the plaintext**, analysed in [DS18, RZ18].
- **[ADL17]** Propose **GCM-RUP**, an AEAD scheme realized from GCM components that is **secure** under the **release of unverified plaintext**.
- **[ADL17]** also proposed GCM-RUP for **use in Tor**, claiming that a **WITC is overkill**, an OE scheme is described but **without any formal analysis**.

Onion Encryption from GCM-RUP

- In [ADL17] the Onion Encryption scheme was **underspecified, no integrity, no leaky pipes, no ciphertext chaining, and not forward secure**.
- [Prop 295-Jul 2018] Updated version **added leaky pipes and integrity, but not IND-CPA secure, still no ciphertext chaining and forward security**.
- [Prop 295-Mar 2019] Heavily revised scheme, made stateful by **feeding back GHASH outputs** (invalidating proofs) and **extending GCM-RUP to full WITC (STPRP)**, forfeiting the performance gains originally sought!

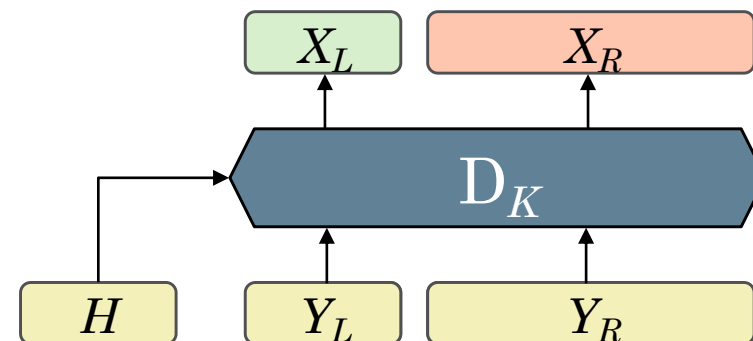
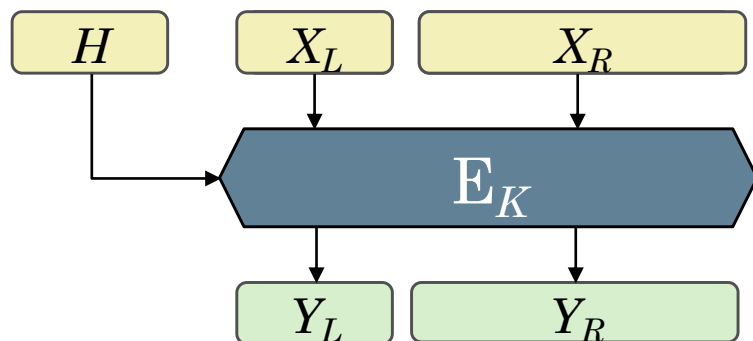
The Design of CGO

The Path Towards CGO



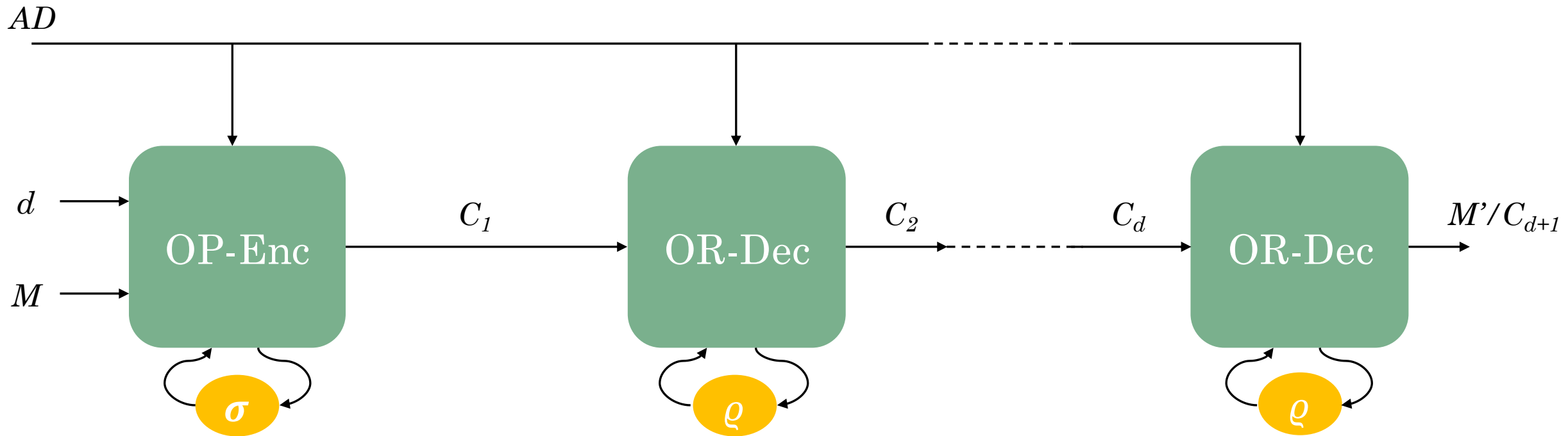
- We extracted a **length-preserving primitive** from GCM-RUP, which can be abstracted as a **Rugged PRP**.
- Augmented the **Rugged PRP** construction with a **key update** functionality to achieve **forward security**.
- A new formal treatment of OE extending prior ones [DS18, RZ18] with **Leaky Pipes, Associated Data, Forward Security**.
- We build CGO in a modular fashion from any **Updateable Rugged PRP** and **prove it secure** in our new model.

Main Building Block: Rugged PRPs



- A Rugged PRP is a **tweakable cipher** over a **split domain**: $\{0,1\}^n \times \{0,1\}^*$, $n = 128$.
- **Intermediate Security**, between TPRP and TSPRP security.
- **Very Informally**: Enciphering is **pseudorandom**, whereas Deciphering only provides **partial non-malleability**.

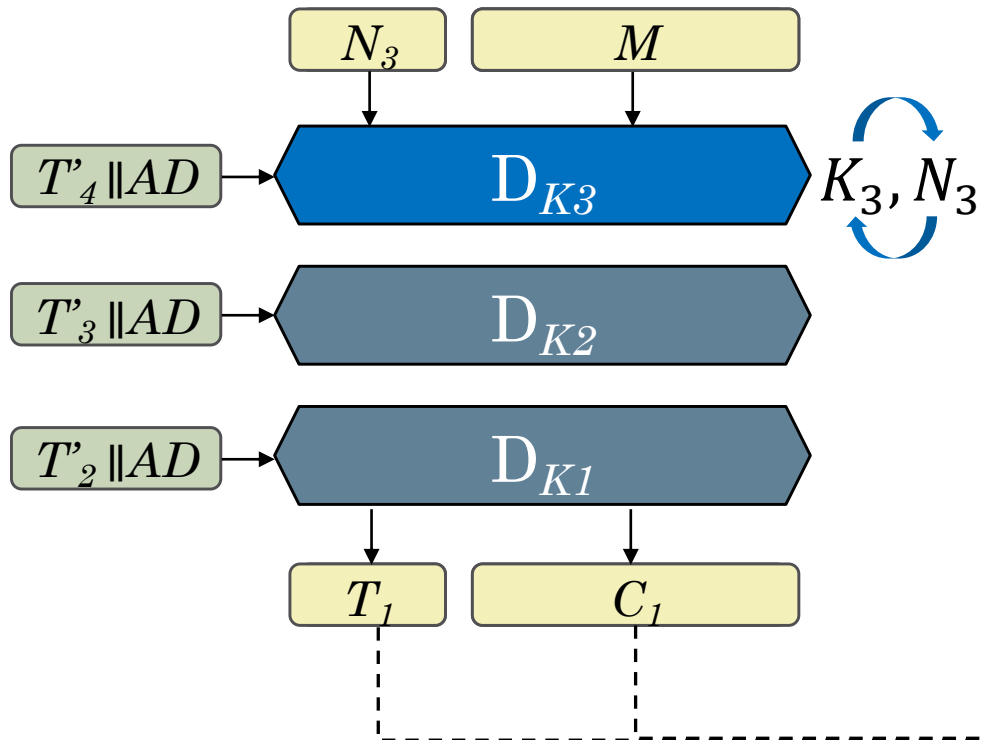
Syntax of Onion Encryption (Forward Direction)



- **Note:** decryption cannot detect invalid ciphertexts – **does not return $\perp$** .
- Hence the need to **chain ciphertexts** - if a **ciphertext is tampered** subsequent ciphertexts **decrypt to garbage**.

CGO (High-Level): Forward Direction

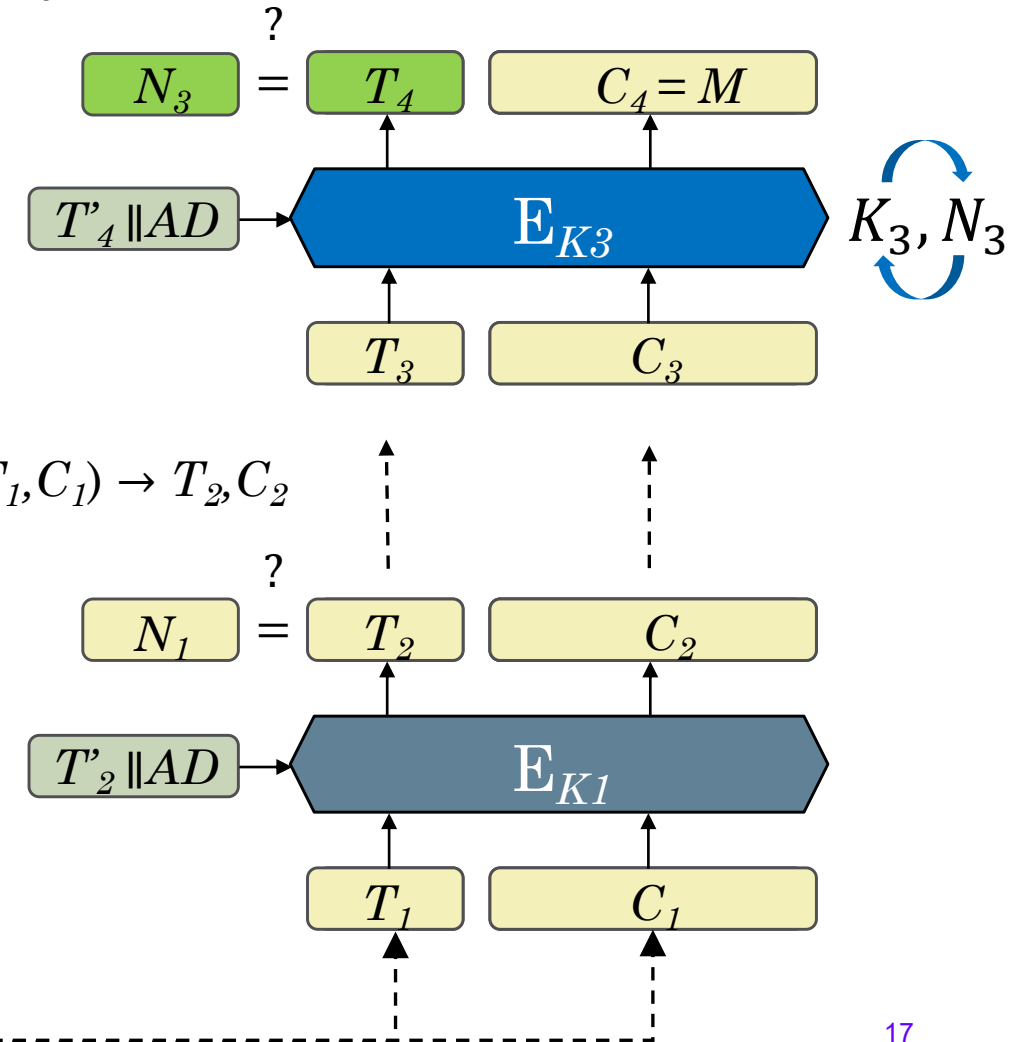
OP-Enc($\sigma, d=3, AD, M$) $\rightarrow T_1, C_1$ $\sigma = [(K_i, N_i, T'_{i+1})]$



OR-Dec(ρ_3, AD, T_3, C_3) $\rightarrow M$

$\rho_i = (K_i, N_i, T'_{i+1})$

OR-Dec(ρ_1, AD, T_1, C_1) $\rightarrow T_2, C_2$

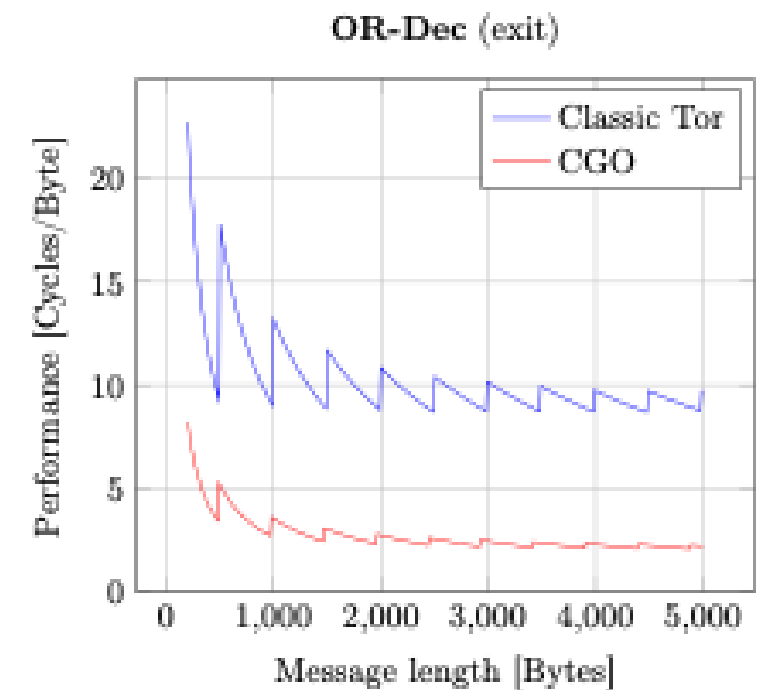
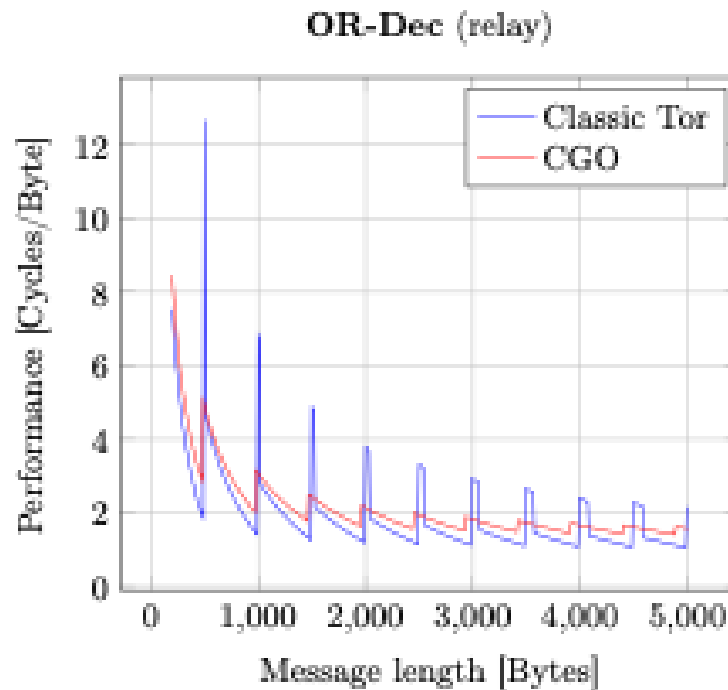
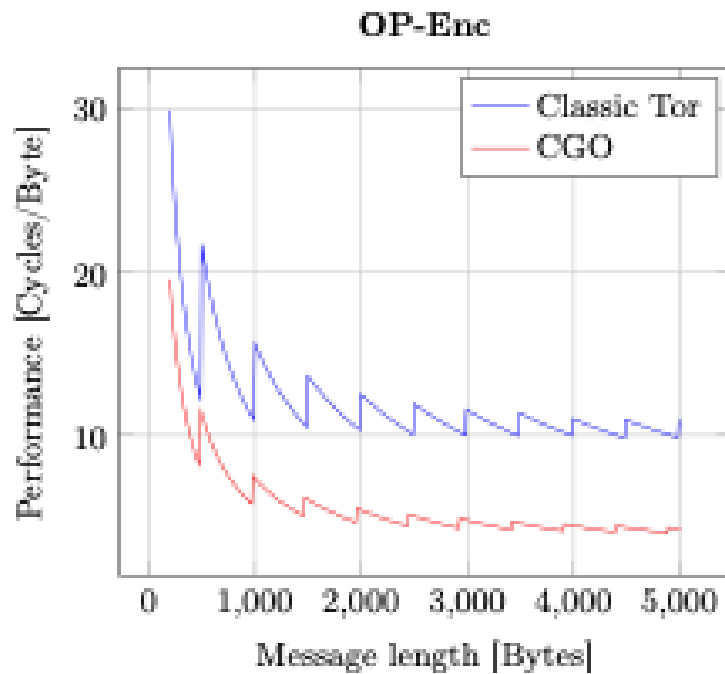


High-Level Design Rationale



- **Non-malleability only where needed**—at the Onion Routers through RPRP enciphering, it is not needed at the Onion Proxy.
- **The nonce** serves to : 1) **diversify** ciphertexts 2) not repeat **left RPRP inputs** 3) provide **integrity** 4) support **leaky pipes**.
- The **tag** acts as a **compressed digest** of the ciphertext, allowing **efficient ciphertext chaining**—preventing recovery from invalid ciphertexts.
- **Associated Data**, prevents tagging in the headers (**CMU 2014 Attack**).

CGO Performance

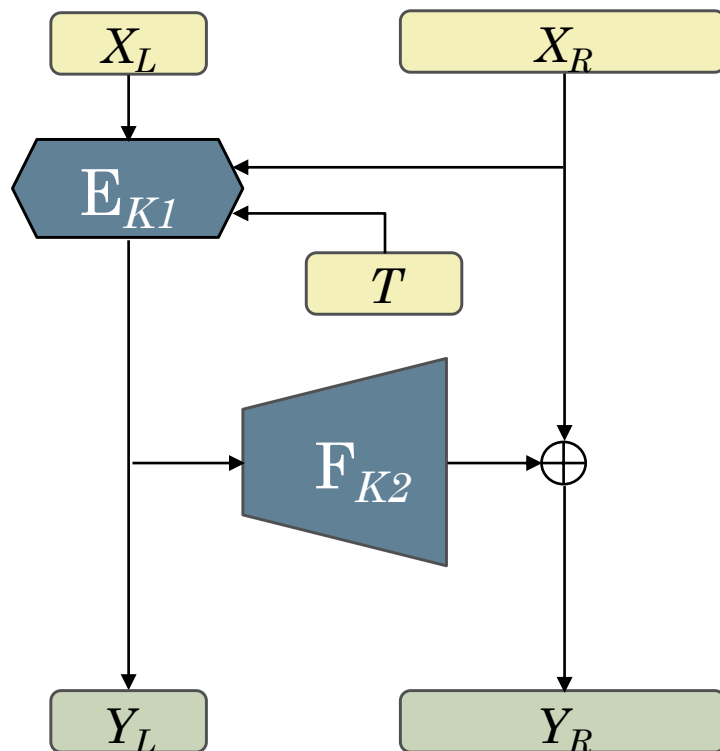


Summary

Summary

- Compared to Legacy Tor, CGO protects against **tagging attacks**, provides **stronger integrity** covering **associated data**, and adds **Forward Security**.
- All while maintaining a comparable (arguably better) **performance tradeoff**.
- CGO's **modular design** means that the **RPRP construction** can be easily replaced if needed, say with a **BBB construction** or even a **STPRP**!
- CGO is currently being integrated in **Arti**—Tor's upcoming Rust implementation.
- [iacr eprint: /2025/583 2025/2017](#)

Unilaterally-Protected IV



- **UIV** is obtained by removing the **third layer** from **PIV** and is secure as an **RPRP**.
- It can be instantiated with **GCM components** leading to a **performance** similar to GCM-SIV.
- It is closely related to **GCM-RUP** [ADL17].